AutoConcourse: A Computational Pipeline for Automatic Generation of Q-Concourse Statements

(Work in Progress)

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June 3, 2026

Abstract

Q-methodology studies subjectivity by having participants sort statements drawn from a *concourse*, i.e. a set of statements selected to be representative for the public discourse on a given issue. The validity of the methodologies findings depends on the coverage, clarity, and balance of this set of statements, but concourse construction remains slow, costly, and difficult to standardize across topics and languages, making Q-methodology hard and expansive to use. We present AutoConcourse, a steerable retrieval-and-generation pipeline that assembles issue-scoped corpora, organizes them via anchor definitions and multilingual sentence embeddings, and synthesizes concise neutral statements with large language models (LLMs) under human control. This paper reports the pipeline in its current state, implemented and demonstrated on the generation of a Deliberative Reason Index (DRI) survey — a specific application of Q-methodology — for a Swiss citizens' assembly on healthcare costs, with an auxiliary comparison to a human-authored DRI survey on food policy. We explicitly flag two open gaps that stand between the current implementation and the generalizable concourse-generation tool promised by the pipeline's framing: anchors currently depend on an externally authored policy analysis rather than being derived autonomously from the corpus, and validation so far consists of a single qualitative comparison performed by the author. The intent of this submission is to invite feedback from the CompText community on methodological choices, validation protocols, and pipeline generalization.

Keywords Q-methodology; concourse; statement generation; semantic retrieval; multilingual sentence embeddings; large language models; deliberative reason index.

1 Introduction

Q-methodology (Stephenson, 1953) is a mixed qualitative–quantitative approach that studies structured subjectivity by correlating persons (Q-sorts) rather than variables. A study begins by constructing a *concourse*: a set of statements representative for the public discourse surrounding an issue, from which a smaller *Q-sample* of 30–60 statements is selected for participants to rank. The validity of the extracted factors depends directly on the breadth, neutrality, and balance of this statement pool. Building a concourse well is therefore central to every Q study, yet the process remains labor-intensive, idiosyncratic across research teams, and hard to scale, even more so in multilingual settings or when a study must be deployed on a short timeline.

NLP technologies, including multilingual sentence embeddings and large language models (LLMs) make it plausible to automate most of this process. Retrieval-augmented generation (RAG) (Lewis et al., 2021; Gao et al., 2024) provides a template for combining semantic retrieval

with constrained text generation and anchor-based retrieval (Aceves and Evans, 2023) allows for much more fine-grained retrieval than unsupervised clustering when the researcher already has structured hypotheses about the relevant subtopics. Combined, these techniques form a pipeline in which (i) a large corpus is assembled for the given issue, (ii) a set of anchor definitions describing subtopics, frames, and stances is embedded alongside the corpus, (iii) the most anchor-similar paragraphs are retrieved, and (iv) an LLM rewrites them as short, neutral, single-claim statements suitable for Q-sorting.

Contribution. We present AutoConcourse, a modular pipeline that implements this idea end-to-end and demonstrate it on the generation of a Deliberative Reason Index (DRI) survey (Niemeyer and Veri, 2022) for a Swiss citizens' assembly on healthcare costs Kübler and Stojanovic (2022). We contribute: (1) the pipeline architecture with documented prompts and configurations, (2) a supervised alternative to unsupervised discourse clustering, including a *negative result* showing that HDBSCAN (McInnes et al., 2017) and BERTopic (Grootendorst, 2022) fail to produce argumentative clusters on our corpus, (3) an application to a live Swiss citizens' assembly, and (4) a preliminary validation comparing the pipeline's output to a human-authored DRI survey on food policy.

Scope of this paper. The abstract of this submission frames AutoConcourse as a general tool for Q-concourse generation. The implementation and evidence reported here cover the specific case of DRI, which is a well-scoped Q-methodology variant (Niemeyer and Veri, 2022). Generalization to the full range of Q studies is an explicit goal, not a demonstrated property. We flag two gaps that stand between what we have built and what the general framing claims:

- **G1 (Anchor Autonomy Gap).** Our anchor definitions are produced by an LLM *informed by an externally authored policy report*. The pipeline does not yet derive anchors from minimal input from the researcher or even the corpus itself. This limits generalizability because many use-cases of a Q-concourse tool will not have an analogous report available.
- **G2 (Validation Gap).** Our primary validation is a single qualitative comparison against one human-authored DRI survey, carried out by the author. This is evidence of plausibility, not of pipeline quality in the statistical sense.

In this paper, we treat these gaps as the central open problems and describe concrete next steps in Section 8. Our purpose at CompText is to present a system currently under development, report what we have learned, and invite feedback on validation design and generalization strategy.

The rest of the paper proceeds as follows. Section 2 summarizes Q-methodology and locates DRI within it. Section 3 reviews related computational work. Section 4 details the pipeline. Section 5 describes the healthcare-costs case study. Section 6 reports the food-policy comparison. Section 7 returns to the two gaps and their implications. Section 8 sketches the path forward.

2 Background: Q-Methodology and the Concourse

2.1 Q-Methodology in Brief

A Q study proceeds in four steps. First, the researcher demarcates a topic and assembles a *concourse* of candidate expressions surrounding it. Second, they select and phrase a *Q-sample* of $N \approx 30$–60 short declarative statements that collectively represent the major subtopics, frames, stakeholder positions, and stance polarities present in the discourse. Third, a purposefully chosen *P-sample* of participants sorts these statements on a bipolar forced-distribution grid (e.g. most agree to most disagree). Fourth, the resulting by-person correlation matrix is factored, rotated, and interpreted to produce a small set of ideal-typical viewpoints (Brown, 1980; Watts and Stenner, 2012).

Statement-craft norms. Across the methodological literature, good Q-statements are short, declarative, single-claim, neutrally phrased, and comparable in length and reading level. Stance polarity across the set should be approximately balanced and the set should cover substantive facets of the discourse rather than paraphrasing one of them at length (Watts and Stenner, 2012). These norms translate directly into constraints we impose on the LLM at the generation stage.

2.2 The Concourse and Its Pitfalls

The concourse is not analyzed directly, it is a reservoir from which the Q-sample is drawn. In practice, researchers build it from professional media, policy documents, stakeholder materials, and elicited texts such as interviews. Common failure modes are well documented: narrow sourcing that mirrors the researcher's priors; conceptual redundancy that crowds out underrepresented frames, inconsistent register that burdens sorting, and, in multilingual studies, translation drift that subtly shifts stance (Watts and Stenner, 2012). The pipeline we present is designed to address failure modes. Through automatization, every retained statement can carry provenance, and balance across languages, outlets, and stances becomes a tunable quota rather than an accident of collection.

2.3 DRI as a Specific Q Application

The Deliberative Reason Index (Niemeyer and Veri, 2022) is an outcome measure for deliberation that applies Q-methodology to pre- and post-deliberation surveys. It measures *intersubjective consistency*, i.e. the extent to which participants who agree on *considerations* (reasons) also agree on *preferences* (policy options), without requiring unanimity (Dryzek and Niemeyer, 2006; Niemeyer and Dryzek, 2007). A DRI survey thus has two linked components: a set of consideration statements and a set of policy options. Both are expected to cover the relevant facets of the public discourse on the issue, and gaps in either lead to distortions in the correlation structure. This makes DRI a useful test case for concourse automation: it is a real Q instantiation with clear success criteria and, in our case, a concrete downstream deployment (a Swiss citizens' assembly, Section 5).

3 Related Work

Automated assessment of deliberation. Most existing ML work on deliberation targets the Discourse Quality Index rather than outcome measures. DelibAnalysis (Fournier-Tombs and Di Marzo Serugendo, 2019) trains random-forest classifiers on human-annotated corpora, AQuA (Behrendt et al., 2024) uses transformer adapters to score multiple deliberative dimensions and aggregates them against expert weightings, and Falk and Lapesa (2022) compare feature-based and transformer classifiers on deliberative quality labels. These are classification tasks over existing text. The generative task of *producing* a concourse from scratch has, to our knowledge, not been previously addressed with an end-to-end pipeline.

Retrieval-augmented generation. RAG (Lewis et al., 2021) grounds LLM outputs in retrieved documents and has become a standard technique for reducing hallucination (Gao et al., 2024). Our pipeline follows the same logic: the generation step is conditioned on top-K paragraphs retrieved from the issue-scoped corpus, which we view as an approximation of the public sphere rather than a parametric memory.

Anchored embeddings. Aceves and Evans (2023) introduce the use of theory-derived anchor texts as semantic reference points in word-embedding spaces, allowing researchers to probe a

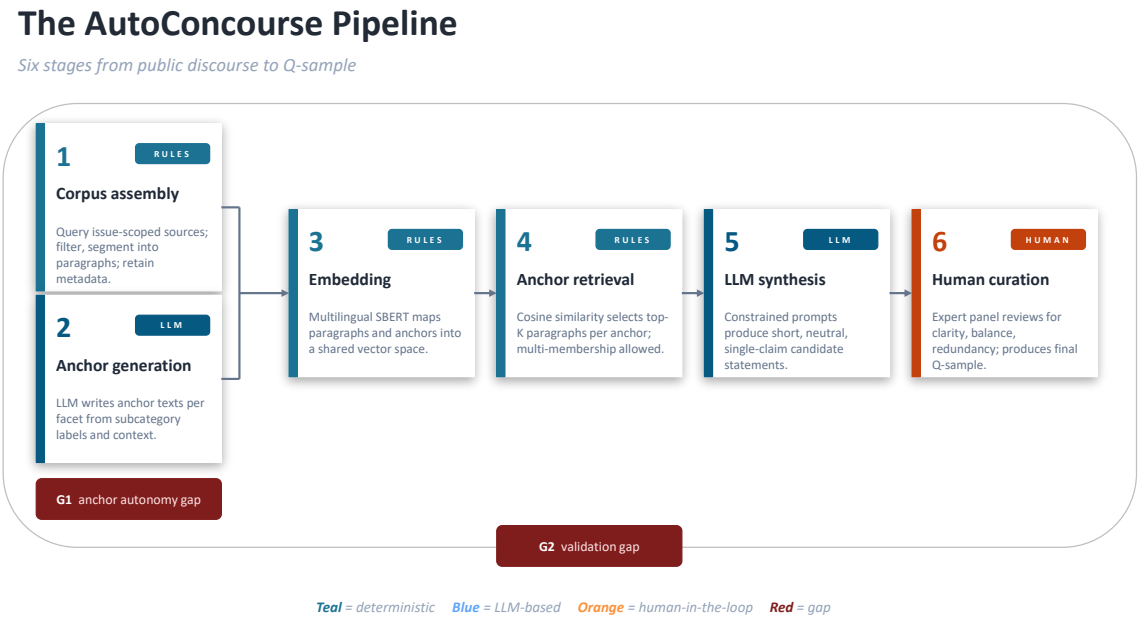


Figure 1: Overview of the AutoConcourse pipeline. Stages 2 and 5 are LLM-based; the remainder are deterministic or human-in-the-loop.

corpus along predefined conceptual dimensions. We adopt this idea at the paragraph level and extend it with multi-membership retrieval (a paragraph can score high on several anchors) and stratification by source ideology.

LLMs for deliberative practice. Tessler et al. (2024) show that an LLM can iteratively draft group-agreeable statements from participant inputs. This complements our work: they generate consensus texts from a discussion, we generate a diverse statement pool from public discourse.

4 Method: The AutoConcourse Pipeline

The pipeline has six stages (Figure 1): (1) corpus assembly, (2) anchor generation, (3) embedding, (4) anchor-based retrieval, (5) constrained LLM synthesis, (6) human curation. Stages 1, 3, 4 are deterministic and auditable, stages 2 and 5 depend on LLM outputs, stage 6 is expert-driven. We describe each below, referring to our pilot as a concrete example application and flagging where gaps **G1** and **G2** apply.

4.1 Corpus Assembly

We treat large multilingual news archives as a tractable proxy for the public discourse on a given issue. For the case study in Section 5, we queried the Swissdox@LiRI database with seventeen keywords related to healthcare costs in German, French, and Italian over the period 2018-01-01 to 2024-08-29, retrieving 444,869 unique articles. Articles were segmented into paragraphs, balancing semantic completeness and embedding granularity, and a second keyword filter removed off-topic paragraphs, leaving us with 738,598 paragraphs. Source, outlet, date, language, and URL were retained as metadata, as was each outlet's political leaning from Udris (2023), coded into a five-point scale (left, left-liberal, centrist, right-liberal, right). This coding allows for stratification of bias in the dataset.

4.2 Anchor Generation (Gap G1)

An anchor is a 3–5-sentence paragraph describing a subtopic that the researcher wants the concourse to cover. In the current implementation, anchors are produced by prompting GPT-4o with (a) a short subcategory label and (b) a pre-existing expert policy analysis as context. For the healthcare-costs case study, the context document was a fifty-page policy-field analysis commissioned from the Interface institute for the organizers of the citizens' assembly (Bischof et al., 2024). Seven subcategories and one overarching anchor were generated in German, French, and Italian each, for $8 \times 3 = 24$ anchor texts.

Why this works and what it costs. Grounding anchors in an expert document yields outputs that reflect domain expertise and ensures that low-frequency but important subtopics, such as hospital financing structures that only specialists discuss in depth, are not missed, which was an explicit requirement for this pilot. The cost is **G1**: the pipeline is not yet self-contained. Applying it to a new topic requires either an existing expert analysis or a researcher willing to write one. For the general Q-concourse tool promised by the abstract, anchors would need to be derived from the corpus itself (for example, by clustering over the corpus and summarizing the clusters). We discuss candidate strategies in Section 8.

4.3 Embedding and Dimensionality Reduction

Paragraphs and anchors are embedded into a shared 384-dimensional space using `paraphrase-multilingual-MiniLM-L12-v2` (Reimers and Gurevych, 2019). The model is multilingual, has a strong price–performance ratio for semantic retrieval on well-edited text (Mahajan et al., 2025), and is small enough to embed ∼740k paragraphs on commodity hardware.

We then apply UMAP (McInnes et al., 2020) to reduce dimensionality to 50 with cosine distance, `n_neighbors`=30, and `min_dist`=0, which empirically preserved the local structure useful for anchor-based retrieval. With contemporary hardware, UMAP is not strictly necessary for the final retrieval step but proved helpful for the exploratory analyses reported in Section 4.7.

4.4 Anchor-Based Retrieval

For each anchor a and paragraph p, we compute $\text{sim}(a, p) = \cos(e_a, e_p)$ on the reduced embeddings. Paragraphs are assigned to all anchors on which they score highly (multi-membership) as real public-discourse paragraphs often touch several subtopics simultaneously.

A practical question is *how many* top paragraphs to retrieve per anchor. In the case study we found that the top 50,000 paragraphs by anchor scored above 0.92 but overlapped approximately 80% across anchors, which would have caused the LLM to generate near-identical statements in different categories (Figure 2). Tightening to the top 500 per anchor reduced average cross-anchor overlap to 14.5%, at the cost of some paragraphs with cross-cutting relevance. This trade-off between anchor-purity and coverage can be considered a tunable property of the pipeline rather than a one-off choice.

4.5 Constrained LLM Synthesis

For each (anchor × ideological bucket) combination, the top 500 paragraphs are passed as context to one of two GPT-4o assistants configured via the OpenAI Assistants API:

- The *Considerations Assistant* produces five candidate consideration statements per combination. Its system prompt defines the DRI concept, enumerates style constraints (single-claim, neutral, concise), attaches ∼650 example statements from previous DRI studies as a reference vector store, and returns structured JSON.

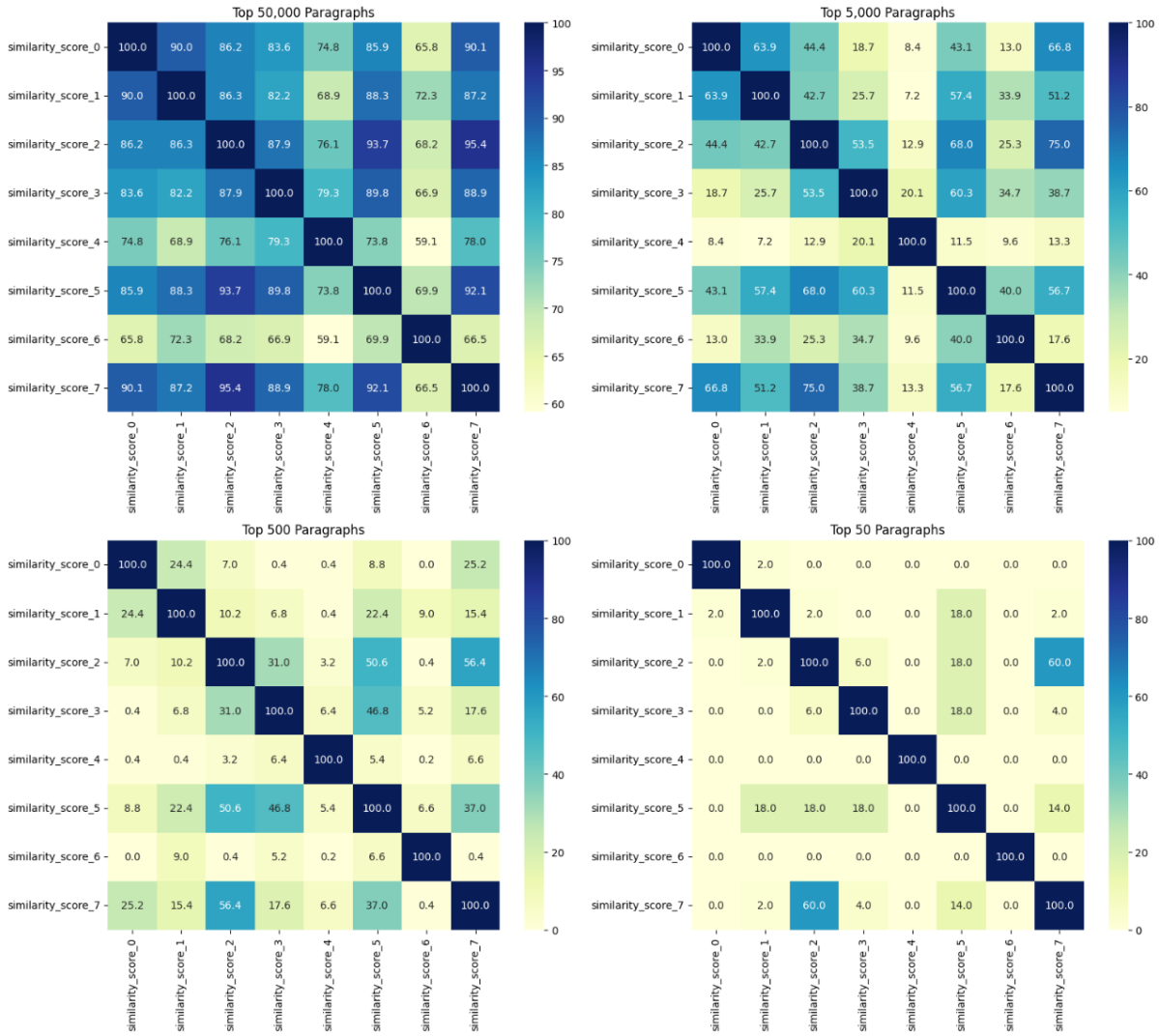


Figure 2: Pairwise overlap of retrieved paragraphs across anchors at four retrieval depths. The top-500 regime substantially reduces redundancy across anchors while retaining enough material for synthesis.

- The *Policy Assistant* produces five candidate policy options per combination, with an analogous prompt and ∼100 prior policy options as its reference set.

Stratification by ideology matters twice: it gives under-covered perspectives a guaranteed slot in the candidate pool, and it lets the human curator see, explicitly, whether a statement expresses a left-coded or right-coded framing before deciding to keep it.

For the healthcare case study, this procedure produced 200 consideration statements and 25 policy options in total across the eight anchors and five ideological buckets.

4.6 Human Curation

A panel, in our case the scientific advisory board of the citizens' assembly, reviews the candidates for redundancy, clarity, and balance, selecting a final pool that fits the citizens'-assembly timing constraints. In the healthcare case the pool was cut to 28 items.

4.7 A Useful Negative Result: Unsupervised Clustering

Before settling on anchor-based retrieval, we tried the more obvious route: cluster the paragraph embeddings with HDBSCAN (McInnes et al., 2017) and summarize clusters with BERTopic (Grootendorst, 2022). However, the resulting clusters were dominated by named entities — pharmaceutical companies, federal offices, individual politicians — rather than argumentative themes. Cluster centroids did not read as coherent stances. Quantitatively, when we projected clusters onto our anchor scores, the maximum gap between anchor similarities within a cluster was only 0.185 among the top-20 clusters, and these gaps sat at relatively low absolute similarity (≤ 0.6). A search for clusters with one high anchor score and all others below a moderate threshold yielded only two items. The conclusion is not that HDBSCAN or BERTopic are weak tools in general, but that this unsupervised clustering approach did not base clusters on actual sub-discourses but rather on keywords.

5 Case Study: DRI for Swiss Healthcare Costs

The pipeline was developed alongside the *Bevölkerungsrat 2025*, a citizens' assembly on healthcare costs in Switzerland coordinated by the Zentrum für Demokratie Aarau in partnership with the Universities of Zurich and Geneva, running from November 2024 to March 2025 with 100 participants drawn by stratified sortition from an initial contact pool of 22,000. Healthcare costs are the top concern in recent Swiss opinion data (Golder et al., 2024).

Application. The pipeline was used to generate the DRI survey administered to participants before and after deliberation. Anchors were constructed around seven substantive subcategories identified by the organizers (health system reforms, healthcare utilization, hospital planning and financing, compulsory insurance design, health prevention incentives, health service coordination, household financial burden) plus one overarching anchor focusing on high health costs in general. The resulting 28-item consideration set and 8-item policy-options set were used in the live survey.

Pre-deliberation DRI. The pre-deliberation DRI score for the assembly was 0.16, which is a plausible baseline level of intersubjective consistency for a randomly sampled population before deliberation. Post-deliberation indicated a small gain in DRI to 0.19. However, for this paper, what matters is that the survey instrument was deployed, participants completed it, and the resulting score was interpretable and in the expected range — *not* that the DRI score itself validates the pipeline.

A second-order signal. Veri (2025) applied a multifactor analysis to the pre-deliberation responses as part of a separate methodological study aimed at shortening the survey. The analysis yielded eight distinct factors with eigenvalues above one, suggesting that the statement set captures at least that many independent dimensions of the discourse — roughly matching the number of anchors we constructed. This is consistent with the pipeline covering the intended facets, though it is not proof as eight factors could also emerge from a differently structured but coincidentally multi-dimensional pool.

6 Preliminary Validation: Food Policy (Gap G2)

Because the healthcare case has no human-authored counterpart, we validated against a second topic — Swiss food policy — where a human-crafted DRI survey was available. Mathys (2024)

constructed a DRI survey for the Swiss Citizens' Assembly for Food Policy (2022) by reconstructing public debate around the Swiss food system, drawing on themes from recent popular initiatives and opinion surveys.

Setup. We ran AutoConcourse on food policy using a keyword-search Swissdox query yielding 393,258 paragraphs after filtering, anchors generated from the expert handouts used at the assembly plus Mathys's own context chapter, and the smaller, cheaper GPT-4o-mini for synthesis.

Procedure. For each human-authored DRI item in Mathys (2024)'s survey, we computed cosine similarity to every pipeline-generated item and reviewed the five closest matches, coding each as a good match, a good-to-okay match, an okay match, or no match. This was done by the author and is the heart of **G2**: a single-rater qualitative review with no inter-annotator agreement check.

Results. Of 41 human-authored consideration statements, the pipeline produced a good match for 14, a good-to-okay match for 3, and an okay match for 3 (20 of 41, ≈49%). Of 8 policy preferences, 5 were good matches and 2 acceptable (7 of 8, ≈88%).

Among the 21 unmatched considerations, 12 were phrased at a high level of generality (e.g. first-person reflective claims about personal agency or ethics) or carried explicitly pessimistic framings. The pipeline's prompts and the underlying data did not produce these styles reliably. Restricting to the remaining 29 substantively specific statements, the pipeline matched 19 (≈66%) and all 7 non-general policy preferences (100%). A concrete example: the human-authored statement *"agricultural subsidies should promote sustainable, pesticide- and antibiotic-free production"* was matched by a pipeline statement advocating prioritization of ecological-farming investments over conventional-agriculture subsidies.

Among the remaining unmatched items were fringe-view statements such as the claim that global warming might *benefit* food production globally — a view rarely expressed in mainstream Swiss media and therefore absent from the corpus. This is a real limitation of any news-corpus-based approach and not solvable by pipeline changes alone.

Validation Frame. These numbers support the claim that the pipeline can recover a substantial fraction of human-authored DRI content, particularly for substantively specific items, and that it does so in a second domain without domain-specific tuning. They do not support claims about consistency across raters, reproducibility across runs, or performance on topics with weaker media coverage. A full validation study would of course need multiple raters, multiple target topics, and comparison of factor-analytic outputs from pilot Q-sorts — not just surface matching of statements.

7 Limitations and the Two Gaps

We return to **G1** and **G2** and list their downstream implications, along with other limitations of the current work.

G1: Anchor autonomy. The current pipeline requires a pre-existing expert analysis for anchor generation. This is acceptable when such an analysis exists (e.g. the Interface report for the *Bevölkerungsrat 2025*, or Mathys's contextual chapter for food policy), but it means AutoConcourse is not drop-in usable for arbitrary Q studies. The pipeline is, strictly speaking, *anchor-conditional*, meaning it automates everything from anchor to concourse, but not the anchors themselves. Generating the anchors, however, still requires much effort, motivating further improvements on the pipeline in this aspect.

G2: Validation breadth. Our validation consists of a single comparison study, judged qualitatively by the author. This is suggestive but cannot distinguish pipeline quality from author-rater bias, and tells us little about run-to-run stability. A credible validation design — which we do not claim to have executed — might involve multiple raters with blinded origin labels, multiple target topics, formal agreement statistics, and ideally a pilot Q-sort comparing factor structures from pipeline-generated vs. human-authored Q-samples. Possibly, more automated validation approaches could justify similar plausibility claims with less effort involved.

Media-coverage dependency. The pipeline inherits whatever is in the corpus. Mainstream news underrepresents minority and fringe views (Buyens and Aelst, 2021), and generating for discourses specific to a country with a weaker or more captured media landscape would produce lower-quality concourses even with an otherwise identical pipeline.

LLM-specific concerns. GPT-4o is closed-weights, so we cannot audit what material from the 500 retrieved paragraphs is actually used. Output is probabilistic, so repeated runs yield different candidate pools. Output style reflects the model's own distribution, which in our experience defaults toward neutral and mildly positive framings — a plausible explanation for the missing pessimistic statements in the food-policy comparison.

Generalization beyond DRI. DRI is a narrow Q variant with explicit considerations and policy options. Broader Q applications involve less-structured statement types (e.g. experiential or moral claims) and different balance norms. The pipeline should transfer, but we do not claim to have shown that it does.

8 Path to Generalizability

Closing **G1** and **G2** is the work that would move AutoConcourse from "works for DRI on healthcare in Switzerland, given an expert report" to a generally applicable method for concourse generation. We outline concrete next steps.

Closing G1: corpus-derived anchors. As of now, the pipeline requires a thorough discourse analysis to identify relevant subtopics, informing the anchor generation. Ideally, this aspect could be simplified to streamline the concourse generation process. There are several approaches that come to mind here: (a) try again to automate the identification of subtopics based on a combination of NLP methodologies, (b) in a less supervisable fashion, it might be possible to extract subtopics from the corpus with the help of LLMs, (c) topics might also be identified from stakeholder documents (party platforms, NGO briefs, government consultations) which are typically available even when a bespoke expert report is not. However, the latter approach would be prone to biases in the stakeholder documents. Each approach could be combined with expert review when available, preserving the current pipeline's auditability. As this automatization of this step is still in the making, any ideas and feedback would be highly appreciated.

Closing G2: a proper validation study. In order to validate the pipeline as universally applicable to concourse generation, it needs to be validated thoroughly. A possible validation approach could look as follows: (i) select ≥ 5 diverse issues where human-authored Q-samples exist for benchmarking and generate a concourse, (ii) get participants to fill in both human-authored and automatically generated Q-survey, (iii) compare similarity of the results. Alternatively, or additionally, it would be possible to get Q-methodology experts to compare the surveys for sufficient similarity. However, the validation approach has not yet been decided upon and any feedback would be highly appreciated.

Beyond these gaps. Longer-term directions include integration of social-media data to capture registers absent from news, use of open-weights models to enable auditing of retrieval-to-generation traces, output sampling across multiple LLM runs with aggregation by frequency for higher accuracy, and explicit stance-control prompting to recover the missing pessimistic and highly generalized statement styles seen in our validation.

9 Conclusion

We presented AutoConcourse, a computational pipeline that assembles an issue-scoped multilingual corpus, organizes it via theory-guided anchors and sentence embeddings, and uses constrained LLM prompting to synthesize short neutral statements suitable for Q-methodology. The pipeline has been deployed in a real Swiss citizens' assembly on healthcare costs and has been compared, in preliminary form, against a human-authored DRI survey on food policy, recovering roughly half of all considerations and most non-general policy options.

We have been explicit about two gaps between the current implementation and the general Q-concourse tool promised by the abstract. Anchor generation currently depends on an externally authored expert document, and validation consists of a single qualitative comparison by the author. These are the concrete directions in which the work needs to grow, and they are what we would most appreciate feedback on from the CompText community. We believe the pipeline in its present form is a useful demonstration that the task is tractable but we do not believe it is yet a finished tool, and this paper is written in that spirit.

Reproducibility

Code and configuration templates are available at `https://github.com/mooorice/automated-DRI-statement-`
Data are not redistributed due to licensing restrictions on the source news corpus, references and hashed identifiers are retained in the repository.

Acknowledgments

This research was supported by the Horizon Europe project *AI4Deliberation* (European Commission, 2024) and received institutional funding from the Department of Political Science (IPZ) at the University of Zurich and ETH Zurich. We gratefully acknowledge this support.

We further thank the organizers of the Bevölkerungsrat 2025 (Kübler and Stojanovic, 2022) for their support during the piloting phase.

Generative AI tools were used during the coding process to assist with code writing and debugging and during the writing process to improve phrasing and clarity of the manuscript. All AI-assisted outputs were reviewed, tested, and edited by the author, who takes full responsibility for the content, analyses, and conclusions presented in this paper.

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